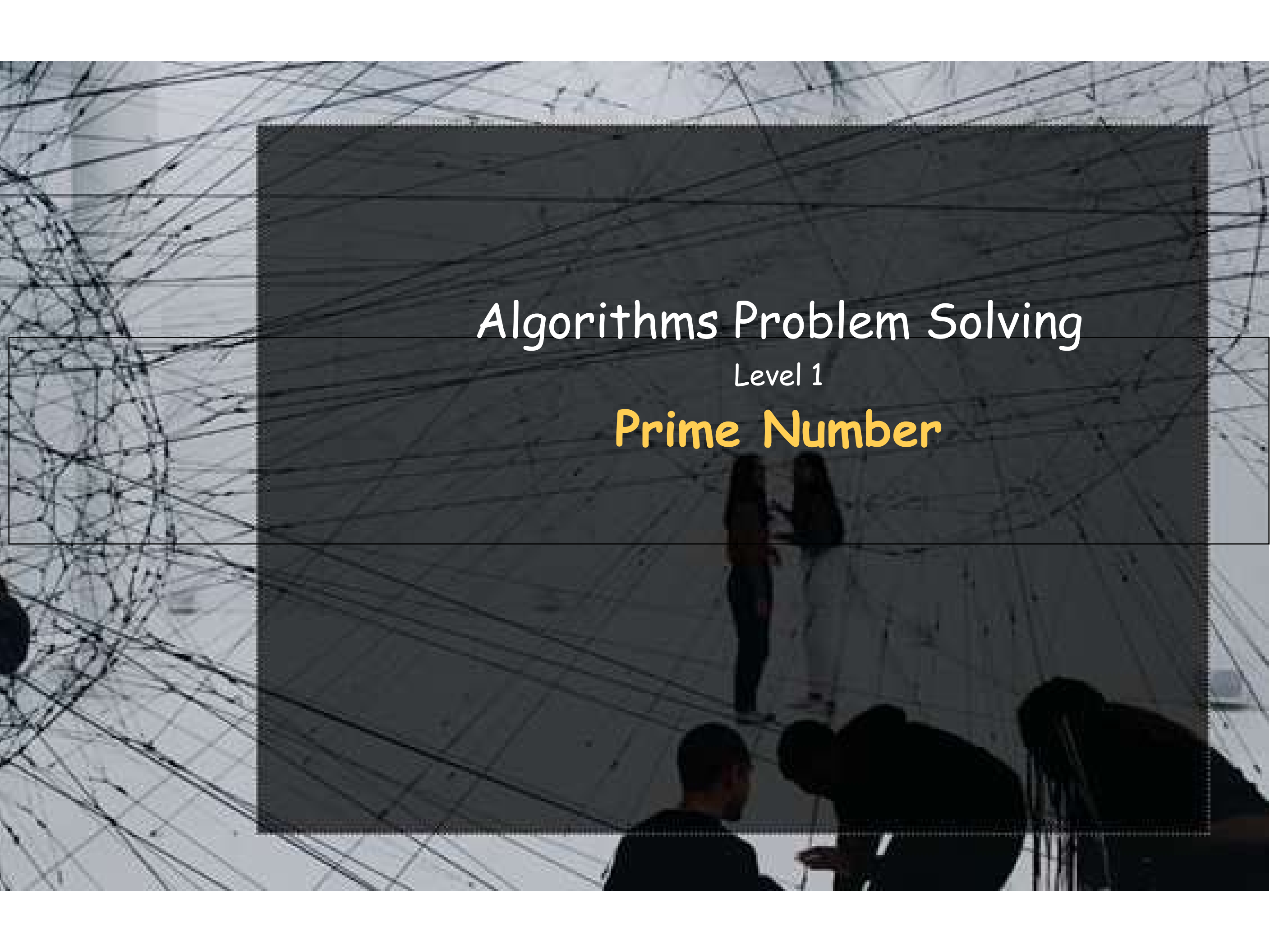


Algorithms Problem Solving

Level 1

Prime Number

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the foreground, the silhouettes of several people are visible, some standing and some sitting, suggesting a collaborative workspace or a public area. The overall tone is professional and tech-oriented.

Algorithms Problem Solving

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Definition

- ◆ This program reads a positive number from the user and checks whether it is a prime number or not.
- ◆ A prime number is a number greater than 1 that is divisible only by 1 and itself.
- ◆ The program uses an enum, functions, a do-while loop, a for loop, and the modulo `%` operator.

Explanation

◆ `enPrimeOrNot` is an enum used to represent two possible results:

`Prime`

`NotPrime`

◆ `readNumber()` keeps asking the user for a number until a positive number is entered.

◆ `checkPrime()` starts checking divisibility from 2.

◆ The expression ``numb % i == 0`` checks whether `numb` is evenly divisible by `i`.

◆ If a divisor is found, the function immediately returns `NotPrime`.

◆ If no divisor is found, the function returns `Prime`.

◆ The loop checks up to ``numb / 2`` because a number greater than half of `numb` cannot divide `numb` evenly, except `numb` itself.

◆ `printResult()` displays the number and whether it is prime.

☐ Steps in main()

- 1☐ Read a positive number from the user.
- 2☐ If the number is less than 2, classify it as NotPrime.
- 3☐ Start checking divisors from 2.
- 4☐ Check whether ``numb % i == 0``.
- 5☐ If a divisor is found, return NotPrime.
- 6☐ Continue checking until ``numb / 2``.
- 7☐ If no divisor is found, return Prime.
- 8☐ Print the result.

💡 Example

◆ If the user enters:

7

◆ The program checks:

$7 \% 2 \neq 0$

$7 \% 3 \neq 0$

◆ No divisor is found.

➡ Result

7 is a Prime number.

Note

- ◆ Examples of prime numbers:
2, 3, 5, 7, 11, 13
- ◆ Examples of non-prime numbers:
4, 6, 8, 9, 10
- ◆ `2` is the smallest prime number.

⊘ Warning

◆ The program asks for a positive number, but numbers `1` and `0` are not prime. The `checkPrime()` function correctly handles numbers less than 2 as NotPrime.



Comparison

Prime

- ✓ Greater than 1.
- ✓ Has exactly two positive divisors: 1 and itself.
- ✓ No divisor is found between 2 and ``numb / 2``.

NotPrime

- ✓ Less than 2, or
- ✓ Has another divisor besides 1 and itself.
- ✓ ``numb % i == 0`` becomes true for a value of `i`.

✦✦ Summary

- ✓ A prime number is greater than 1 and has only two positive divisors.
- ✓ `%` can be used to check whether one number divides another evenly.
- ✓ The function returns NotPrime as soon as a divisor is found.
- ✓ An enum can represent the possible results clearly.

Thank you 😊